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WATER GUN

Abstract

A water gun is provided. On the water gun, when a liquid pump is turned on, liquid in a liquid storage device is injected into a first water tank through a water inlet hole. The liquid pump is used as a unique power source to synchronously perform two functions: In a first function, the liquid in the first water tank is extracted through the water outlet hole for being pressurized, thus forming a high-pressure jet water flow. In a second function, a dynamic pressure field is built inside the first water tank to force a part of the liquid to flow towards a second water tank through a water replenishment hole. The single-pump double-output design directly replaces a shooting water pump and a mist spray water pump that work independently in a traditional double-pump system, and fundamentally reduces power consumption.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to the field of toys, and in particular, to a water gun.

BACKGROUND

[0002] Water guns can pressurize and pump liquid during operation, so that a particular amount of liquid can obtain kinetic energy and be sprayed out at a speed. In order to improve the realism of the existing water gun, a sprayer mechanism is usually added on the water gun. The sprayer mechanism can spray water mist and illuminate the water mist, thus simulating a gun flame effect.

[0003] However, the sprayer mechanism on the water gun needs to be additionally provided with a mist spray water pump and an atomizing water tank. The mist spray water pump pumps liquid into the atomizing water tank, so that an atomizing member on the atomizing water tank can be in direct contact with the liquid, and then the atomizing member can atomize and spray the liquid. In this way, the water gun needs to be provided with two water pumps: a shooting water pump for shooting liquid, and a mist spray water pump, so that the power consumption increases significantly, causing a short battery life of the product.

SUMMARY

[0004] The present disclosure provides a water gun, which can ensure a battery life while maintaining a mist spray function.

[0005] The present disclosure provides a water gun, including: [0006] a gun body; [0007] a liquid storage device, arranged on the gun body; and [0008] a pumping and atomizing device, comprising a liquid pump, a first water tank, a second water tank, and an atomizing member, wherein the liquid pump is arranged on the gun body; a water inlet hole, a water outlet hole, a water replenishment hole, and a flow-back hole are provided in the first water tank; the water inlet hole is communicated to the liquid storage device; the water outlet hole is communicated to the liquid pump; the water replenishment hole is communicated to the second water tank; a through hole is provided in the second water tank; the through hole is communicated to the atomizing member; the second water tank is communicated to the flow-back hole; [0009] when the liquid pump works, liquid inside the liquid storage device enters the first water tank through the water inlet hole; the liquid inside the first water tank is conveyed into the liquid pump through the water outlet hole to be pressurized and sprayed out, and is also conveyed into the second water tank through the water replenishment hole to be further transported to the atomizing member through the through hole; and excess liquid that has not been atomized inside the second water tank flows back to the first water tank through the flow-back hole.

[0010] Preferably, the pumping and atomizing device further includes a box body and an isolator; the isolator is arranged inside the box body; one side of the isolator is enclosed with an inner wall of the box body to form the first water tank; the other side of the isolator is enclosed with the inner wall of the box body to form the second water tank; [0011] the water inlet hole, the water outlet hole, and the through hole are provided in the box body; and the water replenishment hole and the flow-back hole are provided in the isolator.

[0012] Preferably, the atomizing member includes a fixing plate and an atomizing sheet; the fixing plate is arranged on the box body; the atomizing sheet is arranged on the fixing plate; and the atomizing sheet covers the through hole.

[0013] Preferably, the isolator includes an isolation plate and a guide cylinder; both the isolation plate and the guide cylinder are located inside the box body; the guide cylinder is arranged on the isolation plate; the two sides of the isolation plate are respectively enclosed with the inner wall of the box body to form the first water tank and the second water tank; the water replenishment hole is provided in the isolation plate; [0014] one end of the guide cylinder presses against the inner wall

of the box body, so that the guide cylinder and the inner wall of the box body are jointly enclosed to form a guide channel; the water outlet hole is communicated to the guide channel; a flowing hole is provided in one end of the guide cylinder; the flowing hole is communicated to the first water tank and the guide channel; the flow-back hole is provided in the other end of the guide cylinder; [0015] the guide channel receives, through the flow-back hole, excess liquid that has not been atomized inside the second water tank, and further receives, through the flowing hole, the liquid inside the first water tank.

[0016] Preferably, the water gun further includes a light-emitting device; the light-emitting device comprises a mounting cylinder, a bracket, an LED board, and an LED; [0017] the mounting cylinder is arranged on the gun body; the bracket is arranged inside the mounting cylinder; the LED board is arranged inside the bracket; the LED is arranged on the LED board; [0018] a liquid spraying hole is provided in the bracket; the bracket and an inner wall of the mounting cylinder jointly define a plurality of mist spraying holes; the liquid spraying hole is communicated to the liquid pump; the liquid spraying hole is configured to: receive liquid pumped out by the liquid pump and guide the liquid to be sprayed out; and the mist spraying holes are communicated to the atomizing member.

[0019] Preferably, a position of the flow-back hole is higher than a position of the water replenishment hole.

[0020] Preferably, a hole diameter of the water inlet hole and a hole diameter of the water outlet hole are both greater than a water diameter of the water replenishment hole.

[0021] The hole diameter of the water inlet hole and the hole diameter of the water outlet hole are both greater than a hole diameter of the flow-back hole.

[0022] Preferably, the hole diameter of the water inlet hole is 1.5 mm to 4.5 mm; and/or [0023] the hole diameter of the water inlet hole is 1.5 mm to 4.5 mm; and/or [0024] the hole diameter of the water replenishment hole is 0.8 mm to 2.5 mm; and/or [0025] the hole diameter of the flow-back hole is 0.8 mm to 2.5 mm.

[0026] Preferably, the liquid pump includes a motor, a gear transmission group, a pump cylinder, a piston, an elastic reset member, and a nozzle; [0027] the motor is arranged inside the gun body, and is in intermittent-driving connection to the piston through the gear transmission group; the pump cylinder is arranged on the gun body; the piston is slidably arranged inside the pump cylinder; the elastic reset member is in driving connection to the piston; a pump suction hole and a pump-out hole are provided in the pump cylinder; the pump suction hole is communicated to the water outlet hole; the nozzle is arranged on the pump-out hole; [0028] when the motor drives the piston to move through the gear transmission group, the liquid is sucked in through the pump suction hole, and the piston presses against or drives the elastic reset member to deform, thereby increasing elastic potential energy of the elastic reset member; and the elastic reset member is configured to drive the piston to do reset movement when released, so that the liquid in the pump cylinder flows through the pump out hole to the nozzle for spraying.

[0029] Preferably, the gear transmission group includes a driving gear, a plurality of transmission gears, an intermittent gear, and a rack; [0030] the motor is in driving connection to the driving gear; the transmission gears are meshed in sequence; the driving gear is meshed with one of the transmission gears; the intermittent gear is coaxially connected to another transmission gear; the rack is connected to the piston, and the intermittent gear is intermittently meshed with the rack; [0031] when the intermittent gear is meshed with the rack, the driving gear is driven by the motor to drive the transmission gears to rotate, thereby driving the intermittent gear to rotate and driving the piston to move through the rack; and [0032] when the intermittent gear is separated from the rack, the elastic reset member is released to drive the piston to be reset and move.

[0033] Preferably, a mounting slot is provided in the gun body; a liquid guide head is arranged in the mounting slot; the liquid guide head is communicated to the water inlet hole; [0034] the liquid storage device includes a drum and a connector; a liquid chamber is provided inside the drum; the

connector is arranged on the drum; the connector is communicated to the liquid chamber; and the drum is detachably inserted into the mounting slot, so that the connector is communicated to the liquid guide head.

[0035] The present disclosure has the following beneficial effects:

[0036] A water gun is provided. On the water gun, when a liquid pump is turned on, liquid in a liquid storage device is injected into a first water tank through a water inlet hole. The liquid pump is used as a unique power source to synchronously perform two functions: In a first function, the liquid in the first water tank is extracted through the water outlet hole for being pressurized, thus forming a high-pressure jet water flow. In a second function, a dynamic pressure field is built inside the first water tank to force a part of the liquid to flow towards a second water tank through a water replenishment hole. The single-pump double-output design directly replaces a shooting water pump and a mist spray water pump that work independently in a traditional double-pump system, and fundamentally reduces power consumption.

[0037] In addition, the liquid flowing into the second water tank continuously infiltrates the atomizing member through a through hole under the action of gravity, to implement liquid atomization and spraying. Therefore, when an atomizing demand is suspended, that is, when the atomizing member stops atomizing the liquid, the liquid can still be transported to the second water tank, but excess liquid inside the second water tank can still automatically return to the first water tank through a flow-back hole for spraying.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0038] The exemplary implementations of the present disclosure will be described in more details by combining the accompanying drawings. The above and other objectives, features, and advantages of the present disclosure will become more obvious. In the exemplary implementations of the present disclosure, the same reference numerals generally represent the same components.

[0039] FIG. 1 is a schematic structural diagram of a water gun according to some embodiments of the present disclosure;

[0040] FIG. 2 is a schematic structural diagram of the water gun shown in FIG. 1, viewed in another angle;

[0041] FIG. 3 is an exploded view of a water gun in some embodiments of the present disclosure;

[0042] FIG. 4 is a partially schematic structural diagram of the water gun shown in FIG. 3;

[0043] FIG. 5 is a partially schematic structural diagram of a water gun in some embodiments of the present disclosure;

[0044] FIG. 6 is a schematic structural diagram of a pumping and atomizing device in some embodiments of the present disclosure;

[0045] FIG. 7 is an enlarged view of part A in FIG. 5; and

[0046] FIG. 8 is a schematic diagram of an internal structure of a water gun in some embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0047] The implementations of the present disclosure will be described in more details below with reference to the accompanying drawings. Although the accompanying drawings show the exemplary implementations of the present disclosure, it should be understood that the present disclosure can be implemented in various forms, and should not be limited to the implementations stated herein. Rather, these implementations are provided for understanding the present disclosure more thoroughly and completely, and can completely transfer the scope of the present disclosure to those skilled in the art.

[0048] It should be understood that although various information may be described using terms

such as “first”, “second”, and “third” in the present disclosure, such information should not be limited to these terms. These terms are only used to distinguish information of the same type from each other. For example, without departing from the scope of the present disclosure, the first information can also be referred to as the second information, and similarly, the second information can also be referred to as the first information. Thus, features defined as “first” and “second” explicitly or implicitly include one or more of the features. In the description of the present disclosure, “plurality” means two or more, unless otherwise expressly and specifically defined. [0049] In the descriptions of the present disclosure, it should be understood that orientations or positional relationships indicated by the terms “length”, “width”, “upper”, “lower”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, and the like are orientations or positional relationships as shown in the drawings, and are only for the purpose of facilitating and simplifying the descriptions of the present disclosure instead of indicating or implying that devices or elements indicated must have particular orientations, and be constructed and operated in the particular orientations, so that these terms are not construed as limiting the present disclosure.

[0050] Unless otherwise expressly specified and limited, the terms “mount”, “connect”, “connection”, “fix” the like should be understood in a broad sense, such as, a fixed connection, a detachable connection, an integrated connection, a mechanical connection, an electrical connection, a direct connection, an indirect connection through an intermediate medium, an internal communication of two elements, or interaction between two elements. For those of ordinary skill in the art, the specific meanings of the aforementioned terms in the present disclosure can be understood based on specific conditions.

[0051] FIG. 1 and FIG. 2 show a water gun **10** in some embodiments of the present disclosure. The water gun **10** includes a gun body **1**, a liquid storage device **2**, and a pumping and atomizing device **3**.

[0052] It can be understood that an appearance structure of the gun body **1** can be flexibly set, and the gun body **1** plays a role of mounting and supporting other components. The liquid storage device **2** is arranged on the gun body **1** and is configured to store liquid. The pumping and atomizing device **3** is arranged on the gun body **1**, and is communicated to the liquid storage device **2**. The pumping and atomizing device **3** is configured to pressurize and spray liquid, and can also atomize and spray liquid.

[0053] As shown in FIG. 1 to FIG. 8, the pumping and atomizing device **3** includes a liquid pump **31**, a first water tank **32**, a second water tank **33**, and an atomizing member **34**. The liquid pump **31** is arranged on the gun body **1**. A water inlet hole **321**, a water outlet hole **322**, a water replenishment hole **323**, and a flow-back hole **324** are provided in the first water tank **32**. The water inlet hole **321** is communicated to the liquid storage device **2**. The water outlet hole **322** is communicated to the liquid pump **31**. The water replenishment hole **323** is communicated to the second water tank **33**. A through hole **331** is provided in the second water tank **33**. The through hole **331** is communicated to the atomizing member **34**. The second water tank **33** is communicated to the flow-back hole **324**.

[0054] When the liquid pump **31** works, the liquid inside the liquid storage device **2** enters the first water tank **32** through the water inlet hole **321**. The liquid inside the first water tank **32** is conveyed into the liquid pump **31** through the water outlet hole **322** to be pressurized and sprayed out, and is also conveyed into the second water tank **33** through the water replenishment hole **323** to be further transported to the atomizing member **34** through the through hole **331**. Excess liquid that has not been atomized inside the second water tank **33** flows back to the first water tank **32** through the flow-back hole **324**.

[0055] It can be understood that the liquid pump **31** is arranged on the gun body **1**. The liquid pump **31** is configured to: apply a kinetic pressure to the liquid and output the liquid. The first water tank **32** is communicated to the liquid storage device **2** through the water inlet hole **321** to receive basic

liquid flow. The water outlet hole **322** of the first water tank **32** guides the liquid to the liquid pump **31** for pressurized spraying. The water replenishment hole **323** is communicated to the second water tank **33** to transport liquid for atomization, and the flow-back hole **324** and the second water tank **33** form a closed loop to recycle excess liquid.

[0056] The second water tank **33** is communicated to the atomizing member **34** through the through hole **331**. By the formation of the through hole **331**, the liquid can flow to the atomizing member **34** for atomization. The communication between the through hole **331** and the water replenishment hole **323** achieves liquid diversion, and the communication between the through hole **331** and the flow-back hole **324** ensures the cyclic utilization rate of the liquid. The atomizing member **34** is configured to atomize and spray the liquid transported by the through hole **331**.

[0057] It should be noted that when the liquid pump **31** is turned on, the liquid inside the liquid storage device **2** enters the first water tank **32** through the water inlet hole **321** and is divided into two paths: One path enters the liquid pump **31** through the water outlet hole **322** for pressurization and is then sprayed at a high speed. The other path is conveyed into the second water tank **33** via the water replenishment hole **323**, and is then transported to the atomizing member **34** through the through hole **331** to complete atomization and spraying. Excess liquid that has not been atomized inside the second water tank **33** returns to the first water tank **32** through the flow-back hole **324**, thus forming a dynamic balance.

[0058] It should also be noted that the atomizing member **34** can be configured as various components that can atomize liquid in the existing art.

[0059] As shown in FIG. **3** to FIG. **8**, in some embodiments of the water gun, the pumping and atomizing device **3** further includes a box body **35** and an isolator **36**. The isolator **36** is arranged inside the box body **35**. One side of the isolator **36** is enclosed with an inner wall of the box body **35** to form the first water tank **32**, and the other side of the isolator **36** is enclosed with the inner wall of the box body **35** to form the second water tank **33**.

[0060] The water inlet hole **321**, the water outlet hole **322**, and the through hole **331** are provided in the box body **35**; and the water replenishment hole **323** and the flow-back hole **324** are provided in the isolator **36**.

[0061] It can be understood that the box body **35** is configured to store the liquid transported by the liquid storage device **2**. The isolator **36** is configured to divide an internal space of the box body **35** into two chambers, one of which is the first water tank **32** and the other one of which is the second water tank **33**. Through a nested structure of the isolator **36** and the box body **35**, physical isolation and liquid interaction of the first water tank **32** and the second water tank **33**, a compact structure, and high sealing performance are achieved.

[0062] It should be noted that in this embodiment, the desired connection between the first water tank **32** and the second water tank **33** can be completed by directly providing the water replenishment hole **323** and the flow-back hole **324** in the isolator **36**, so that the water replenishment hole **323** and the flow-back hole **324** can be communicated to the second water tank **33** without pipelines, and liquid supplying to the second water tank and returning of the excess liquid can be completed. This greatly reduces the production costs of the product, reduces the complexity of assembling, and improves the production and assembling efficiency of the product. Meanwhile, it also improves the overall compactness of the product.

[0063] As shown in FIG. **4** to FIG. **8**, in some embodiments of the water gun, the atomizing member **34** includes a fixing plate **341** and an atomizing sheet **342**. The fixing plate is arranged on the box body **35**. The atomizing sheet **342** is arranged on the fixing plate **341**. The atomizing sheet **342** covers the through hole **331**.

[0064] It can be understood that the rigid connection between the fixing plate **341** and the box body **35** ensures accuracy of alignment between the atomizing sheet **342** and the through hole **331**, thus improving atomization stability. The atomizing sheet **342** directly covers the through hole **331**, and the liquid is immediately treated by the atomizing sheet **342** after passing through the through hole

331, thus reducing an atomization delay. Meanwhile, the fixing plate **341** provides a mounting base for the atomizing sheet **342**, making it easy to remove, maintain, or replace the atomizing sheet **342**.

[0065] It should be noted that the atomizing sheet **342** can be configured as a commonly used microporous atomizing sheet in the existing art, or can be configured as another component in the existing art that can receive the liquid and generate and spray water mist.

[0066] As shown in FIG. **4** to FIG. **8**, in some embodiments of the water gun, the isolator **36** includes an isolation plate **361** and a guide cylinder **362**. The isolation plate **361** and the guide cylinder **362** are both located inside the box body **35**, and the guide cylinder **362** is arranged on the isolation plate **361**. The two sides of the isolation plate **361** are respectively enclosed with the inner wall of the box body **35** to form the first water tank **32** and the second water tank **33**. The water replenishment hole **323** is provided in the isolation plate **361**.

[0067] One end of the guide cylinder **362** presses against the inner wall of the box body **35**, so that the guide cylinder **362** and the inner wall of the box body **35** are jointly enclosed to form a guide channel **363**. The water outlet hole **322** is communicated to the guide channel **363**. A flowing hole **3621** is provided in one end of the guide cylinder **362**. The flowing hole **3621** is communicated to the first water tank **32** and the guide channel **363**. The flow-back hole **324** is provided in the other end of the guide cylinder **362**.

[0068] The guide channel **363** receives, through the flow-back hole **324**, excess liquid that has not been atomized inside the second water tank **33**, and further receives, through the flowing hole **3621**, the liquid inside the first water tank **32**.

[0069] It can be understood that the guide cylinder **362** is mounted on the isolation plate **361**, and the guide cylinder **362** and the isolation plate **361** can be configured as being clamped, bonded, screwed, or the like. Certainly, the guide cylinder **362** and the isolation plate **361** can also be configured to be integrally formed and jointly manufactured. The isolation plate **361** divides the interior of the box body **35** into the first water tank **32** and the second water tank **33**, thereby avoiding mixing of the liquid inside the two water tanks (the first water tank and the second water tank) through physical isolation, and achieving controllable liquid interaction through the water replenishment hole **323**. Secondly, by providing the flowing hole **3621** and the flow-back hole **324** in the guide cylinder **362**, flow-back path efficiency of the liquid from the first water tank **32** to the liquid pump **31** and the excess liquid inside the second water tank **33** is optimized.

[0070] It should be noted that in an actual operation process, the guide channel **363** can intensively guide two streams of liquid (pressurized liquid from the first water tank **32** and returned liquid from the second water tank **33**) to the liquid pump **31**. In this way, since the pressurized liquid and the returned liquid flow in through two ends of the guide channel **363** respectively, the interference of liquid turbulence is avoided, thus ensuring the stability of pressurized spraying and liquid circulation.

[0071] As shown in FIG. **1** to FIG. **3**, in some embodiments of the water gun, the water gun further includes a light-emitting device **4**. As shown in FIG. **4** and FIG. **5**, the light-emitting device **4** includes a mounting cylinder **41**, a bracket **42**, a LED board **43**, and an LED **44**.

[0072] The mounting cylinder **41** is arranged on the gun body **1**. The bracket **42** is arranged inside the mounting cylinder **41**. The LED board **43** is arranged inside the bracket **42**. The LED **44** is arranged on the LED board **43**.

[0073] A liquid spraying hole **421** is provided in the bracket **42**. The bracket **42** and an inner wall of the mounting cylinder **41** jointly define a plurality of mist spraying holes **422**. The liquid spraying hole **421** is communicated to the liquid pump **31**. The liquid spraying hole **421** is configured to: receive liquid pumped out by the liquid pump **31** and guide the liquid to be sprayed out. The mist spraying holes **422** are communicated to the atomizing member **34**.

[0074] It can be understood that the mounting cylinder **41** can be configured to be arranged on the gun body **1** in a fastened, bolted, plugged, or integrally formed manner. The bracket **42** is

configured to fix the LED board **43** and separate a flowing passage of the liquid spraying hole **421** from flowing passages of the mist spraying holes **422**. The LED board **43** can be configured with a circuit board that can drive the LED **44** to work to emit light in the existing art. Light emitted by the LED **44** can be emitted into the sprayed water mist. The sprayed water mist refracts the light to enhance a visual effect and simulate a gun flame effect, thereby improving the realism of the product and enriching the function of the product. The liquid spraying hole **421** is configured to directionally guide a water column, which is pressurized by the liquid pump **31**, to be sprayed. The mist spraying holes **422** are configured to allow the water mist to pass through.

[0075] As shown in FIG. **6** and FIG. **8**, in some embodiments of the water gun, a position of the flow-back hole **324** is higher than a position of the water replenishment hole **323**.

[0076] It can be understood that in this embodiment, priority control on liquid supplying is achieved through a spatial height difference, so that the excess liquid can be discharged through the flow-back hole **324** at the higher position only after it ensures that atomization liquid inside the second water tank **33** is sufficient.

[0077] It should be noted that when a liquid level inside the second water tank **33** is lower than the flow-back hole **324**, the water replenishment hole **323** continues to replenish liquid, to preferentially meet a liquid demand of the atomizing member **34**. After the liquid level rises to the height of the flow-back hole **324**, the excess liquid automatically flows back to the first water tank **32** to prevent overflow of the second water tank **33**.

[0078] As shown in FIG. **6** and FIG. **8**, in some embodiments of the water gun, a hole diameter of the water inlet hole **321** and a hole diameter of the water outlet hole **322** are both greater than a water diameter of the water replenishment hole **323**, and the hole diameter of the water inlet hole **321** and the hole diameter of the water outlet hole **322** are both greater than a hole diameter of the flow-back hole **324**.

[0079] It can be understood that the water inlet hole **321** with the large hole diameter quickly replenishes the liquid from the water storage device **2** to the first water tank **32**, thus avoiding idling of the liquid pump **31** due to insufficient liquid supplying. The water outlet hole **322** with the large hole diameter matches a high flow output demand of the liquid pump **31**, to reduce flowing resistance and improving a water column spraying range. The water replenishment hole **323** with the small hole diameter limits a flow velocity of the liquid from the first water tank **32** to the second water tank **33**, to prevent an overload on the atomizing sheet **342** caused by a sudden increase in the liquid level of the second water tank **33**. The flow-back hole **324** with the small hole diameter reduces a flow-back velocity and avoids the interference of a sudden change in pressure on the first water tank **32** with stable operation of the liquid pump **31**.

[0080] More specifically, the hole diameter of the water inlet hole **321** is 1.5 mm to 4.5 mm. The hole diameter of the water inlet hole **322** is 1.5 mm to 4.5 mm. The hole diameter of the water replenishment hole **323** is 0.8 mm to 2.5 mm. The hole diameter of the flow-back hole **324** is 0.8 mm to 2.5 mm.

[0081] As shown in FIG. **4** to FIG. **8**, in some embodiments of the water gun, the liquid pump **31** includes a motor **311**, a gear transmission group **312**, a pump cylinder **313**, a piston **314**, an elastic reset member **315**, and a nozzle **316**.

[0082] The motor **311** is arranged inside the gun body **1**, and is in intermittent-driving connection to the piston **314** through the gear transmission group **312**. The pump cylinder **313** is arranged on the gun body **1**. The piston **314** is slidably arranged inside the pump cylinder **313**. The elastic reset member **315** is in driving connection to the piston **314**. A pump suction hole **3131** and a pump-out hole **3132** are provided in the pump cylinder **313**. The pump suction hole **3131** is communicated to the water outlet hole **322**. The nozzle **316** is arranged on the pump-out hole **3132**.

[0083] When the motor **311** drives the piston **314** to move through the gear transmission group **312**, the liquid is sucked in through the pump suction hole **3131**, and the piston **314** presses against or drives the elastic reset member **315** to deform, thereby increasing elastic potential energy of the

elastic reset member **315**. The elastic reset member **315** is configured to drive the piston **314** to do reset movement when released, so that the liquid in the pump cylinder **313** flows through the pump out hole **3132** to the nozzle **316** for spraying.

[0084] It can be understood that the motor **311** indirectly drives piston **314** to do reciprocating motion through the gear transmission group **312**. The gear transmission group **312** is configured to transmit and convert rotation motion of the motor **311** into linear motion of the piston **314**. The pump cylinder **313** is configured to: accommodate the piston **314** and form a closed chamber. The pump suction hole **3131** and the pump out hole **3132** are respectively configured for liquid suction and discharging. The piston **314** is configured to slide inside the pump cylinder **313**. The piston **314** can transmit the elastic potential energy of the elastic reset member **315** to compress the liquid, and can also generate a negative pressure inside the pump cylinder **313** to pump the liquid. The elastic reset member **315** can deform and store energy when the piston **314** moves, and drive the piston **314** to be reset when released. The nozzle **316** is configured to constrain a liquid spraying direction. Further, the nozzle **316** can be configured in a tapered shape to improve a liquid spraying speed and range.

[0085] As shown in FIG. 3 to FIG. 5, in some embodiments of the water gun, the gear transmission group **312** includes a driving gear **3121**, a plurality of transmission gears **3122**, an intermittent gear **3123**, and a rack **3124**.

[0086] The motor **311** is in driving connection to the driving gear **3121**. The transmission gears **3122** are meshed in sequence. The driving gear **3121** is meshed with one of the transmission gears **3122**. The intermittent gear **3123** is coaxially connected to another transmission gear **3122**. The rack **3124** is connected to the piston **314**, and the intermittent gear **3123** is intermittently meshed with the rack **3124**.

[0087] When the intermittent gear **3123** is meshed with the rack **3124**, the driving gear **3121** is driven by the motor **311** to drive the transmission gears **3122** to rotate, thereby driving the intermittent gear **3123** to rotate and driving the piston **314** to move through the rack **3124**. When the intermittent gear **3123** is separated from the rack **3124**, the elastic reset member is released to drive the piston **314** to be reset and move.

[0088] It can be understood that the driving gear **3121** is connected to an output shaft of the motor **311** to transmit initial rotation power. The plurality of transmission gears **3122** are meshed in sequence to transmit a torque along a predetermined path. The intermittent gear **3123** is periodically meshed with/demeshed from the rack **3124** to control intermittent motion of the piston **314**. The rack **3124** is configured to convert rotation motion of the intermittent gear **3123** into linear displacement of the piston **314**.

[0089] It should be noted that the series arrangement of the plurality of stages of transmission gears **3122** reduces a volume of a transmission system and is suitable for the miniaturization design of the water gun.

[0090] As shown in FIG. 3, in some embodiments of water guns, a mounting slot **11** is provided in the gun body **1**. A liquid guide head **111** is arranged in the mounting slot **11**. The liquid guide head **111** is communicated to the water inlet hole **321**.

[0091] The liquid storage device **2** includes a drum **21** and a connector **22**. A liquid chamber **211** is provided inside the drum **21**. The connector **22** is arranged on the drum **21**. The connector **22** is communicated to the liquid chamber **211**. The drum **21** is detachably inserted into the mounting slot **11**, so that the connector **22** is communicated to the liquid guide head **111**.

[0092] It can be understood that a contour of a slot wall of the mounting slot **11** is adapted to an outer box structure of the drum **21**, so that the drum **21** can be stably assembled inside the mounting slot **11**. After the drum **21** is aligned and assembled into the mounting slot **11**, the connector **22** will be in plugging fit with the liquid guide head **111**, so that the liquid chamber **211** is communicated to the first water tank **32** through the connector **22**, the liquid guide head **111**, and the water inlet hole **321**.

[0093] The present disclosure has the following beneficial effects:

[0094] A water gun is provided. On the water gun, when a liquid pump is turned on, liquid in a liquid storage device is injected into a first water tank through a water inlet hole. The liquid pump is used as a unique power source to synchronously perform two functions: In a first function, the liquid in the first water tank is extracted through the water outlet hole for being pressurized, thus forming a high-pressure jet water flow. In a second function, a dynamic pressure field is built inside the first water tank to force a part of the liquid to flow towards a second water tank through a water replenishment hole. The single-pump double-output design directly replaces a shooting water pump and a mist spray water pump that work independently in a traditional double-pump system, and fundamentally reduces power consumption.

[0095] In addition, the liquid flowing into the second water tank continuously infiltrates the atomizing member through a through hole under the action of gravity, to implement liquid atomization and spraying. Therefore, when an atomizing demand is suspended, that is, when the atomizing member stops atomizing the liquid, the liquid can still be transported to the second water tank, but excess liquid inside the second water tank can still automatically return to the first water tank through a flow-back hole for spraying.

[0096] The schemes of the present disclosure have been described in more details above with reference to the accompanying drawings. In the foregoing embodiments, the description of each embodiment has respective focuses. For a part that is not described in detail in an embodiment, reference may be made to related descriptions in other embodiments. Those skilled in the art should also be aware that the actions and modules involved in this specification may not be necessary for the present disclosure. Furthermore, it can be understood that the steps in the method of the embodiments of the present disclosure can be sequentially adjusted, merged, and deleted according to actual needs. The modules in the device of the embodiments of the present disclosure can be merged, partitioned, and deleted according to actual needs.

[0097] The above has described the various embodiments of the present disclosure. The above explanation is exemplary, not exhaustive, and is not limited to the various embodiments disclosed herein. Many modifications and changes are obvious to those of ordinary skill in the art without deviating from the scope and spirit of the various embodiments described herein. The selection of the terms used herein aims to best explain the principles and practical applications of the various embodiments or improvements to technologies in the market, or to enable other persons of ordinary skill in the art to understand the various embodiments disclosed herein.

Claims

1. A water gun, comprising: a gun body; a liquid storage device, arranged on the gun body; and a pumping and atomizing device, comprising a liquid pump, a first water tank, a second water tank, and an atomizing member, wherein the liquid pump is arranged on the gun body; a water inlet hole, a water outlet hole, a water replenishment hole, and a flow-back hole are provided in the first water tank; the water inlet hole is communicated to the liquid storage device; the water outlet hole is communicated to the liquid pump; the water replenishment hole is communicated to the second water tank; a through hole is provided in the second water tank; the through hole is communicated to the atomizing member; the second water tank is communicated to the flow-back hole; when the liquid pump works, liquid inside the liquid storage device enters the first water tank through the water inlet hole; the liquid inside the first water tank is conveyed into the liquid pump through the water outlet hole to be pressurized and sprayed out, and is also conveyed into the second water tank through the water replenishment hole to be further transported to the atomizing member through the through hole; and excess liquid that has not been atomized inside the second water tank flows back to the first water tank through the flow-back hole.

2. The water gun according to claim 1, wherein the pumping and atomizing device further

comprises a box body and an isolator; the isolator is arranged inside the box body; one side of the isolator is enclosed with an inner wall of the box body to form the first water tank; the other side of the isolator is enclosed with the inner wall of the box body to form the second water tank; the water inlet hole, the water outlet hole, and the through hole are provided in the box body; and the water replenishment hole and the flow-back hole are provided in the isolator.

3. The water gun according to claim 1, wherein the atomizing member comprises a fixing plate and an atomizing sheet; the fixing plate is arranged on the box body; the atomizing sheet is arranged on the fixing plate; and the atomizing sheet covers the through hole.

4. The water gun according to claim 2, wherein the atomizing member comprises a fixing plate and an atomizing sheet; the fixing plate is arranged on the box body; the atomizing sheet is arranged on the fixing plate; and the atomizing sheet covers the through hole.

5. The water gun according to claim 2, wherein the isolator comprises an isolation plate and a guide cylinder; both the isolation plate and the guide cylinder are located inside the box body; the guide cylinder is arranged on the isolation plate; the two sides of the isolation plate are respectively enclosed with the inner wall of the box body to form the first water tank and the second water tank; the water replenishment hole is provided in the isolation plate; one end of the guide cylinder presses against the inner wall of the box body, so that the guide cylinder and the inner wall of the box body are jointly enclosed to form a guide channel; the water outlet hole is communicated to the guide channel; a flowing hole is provided in one end of the guide cylinder; the flowing hole is communicated to the first water tank and the guide channel; the flow-back hole is provided in the other end of the guide cylinder; the guide channel receives, through the flow-back hole, excess liquid that has not been atomized inside the second water tank, and further receives, through the flowing hole, the liquid inside the first water tank.

6. The water gun according to claim 1, wherein the water gun further comprises a light-emitting device; the light-emitting device comprises a mounting cylinder, a bracket, an LED board, and an LED; the mounting cylinder is arranged on the gun body; the bracket is arranged inside the mounting cylinder; the LED board is arranged inside the bracket; the LED is arranged on the LED board; a liquid spraying hole is provided in the bracket; the bracket and an inner wall of the mounting cylinder jointly define a plurality of mist spraying holes; the liquid spraying hole is communicated to the liquid pump; the liquid spraying hole is configured to: receive liquid pumped out by the liquid pump and guide the liquid to be sprayed out; and the mist spraying holes are communicated to the atomizing member.

7. The water gun according to claim 4, wherein the water gun further comprises a light-emitting device; the light-emitting device comprises a mounting cylinder, a bracket, an LED board, and an LED; the mounting cylinder is arranged on the gun body; the bracket is arranged inside the mounting cylinder; the LED board is arranged inside the bracket; the LED is arranged on the LED board; a liquid spraying hole is provided in the bracket; the bracket and an inner wall of the mounting cylinder jointly define a plurality of mist spraying holes; the liquid spraying hole is communicated to the liquid pump; the liquid spraying hole is configured to: receive liquid pumped out by the liquid pump and guide the liquid to be sprayed out; and the mist spraying holes are communicated to the atomizing member.

8. The water gun according to claim 1, wherein a hole diameter of the water inlet hole and a hole diameter of the water outlet hole are both greater than a water diameter of the water replenishment hole, and the hole diameter of the water inlet hole and the hole diameter of the water outlet hole are both greater than a hole diameter of the flow-back hole.

9. The water gun according to claim 4, wherein a hole diameter of the water inlet hole and a hole diameter of the water outlet hole are both greater than a water diameter of the water replenishment hole, and the hole diameter of the water inlet hole and the hole diameter of the water outlet hole are both greater than a hole diameter of the flow-back hole.

10. The water gun according to claim 8, wherein a position of the flow-back hole is higher than a

position of the water replenishment hole; and/or the hole diameter of the water inlet hole is 1.5 mm to 4.5 mm; and/or the hole diameter of the water inlet hole is 1.5 mm to 4.5 mm; and/or the hole diameter of the water replenishment hole is 0.8 mm to 2.5 mm; and/or the hole diameter of the flow-back hole is 0.8 mm to 2.5 mm.

11. The water gun according to claim 9, wherein a position of the flow-back hole is higher than a position of the water replenishment hole; and/or the hole diameter of the water inlet hole is 1.5 mm to 4.5 mm; and/or the hole diameter of the water inlet hole is 1.5 mm to 4.5 mm; and/or the hole diameter of the water replenishment hole is 0.8 mm to 2.5 mm; and/or the hole diameter of the flow-back hole is 0.8 mm to 2.5 mm.

12. The water gun according to claim 1, wherein the liquid pump comprises a motor, a gear transmission group, a pump cylinder, a piston, an elastic reset member, and a nozzle; the motor is arranged inside the gun body, and is in intermittent-driving connection to the piston through the gear transmission group; the pump cylinder is arranged on the gun body; the piston is slidably arranged inside the pump cylinder; the elastic reset member is in driving connection to the piston; a pump suction hole and a pump-out hole are provided in the pump cylinder; the pump suction hole is communicated to the water outlet hole; the nozzle is arranged on the pump-out hole; when the motor drives the piston to move through the gear transmission group, the liquid is sucked in through the pump suction hole, and the piston presses against or drives the elastic reset member to deform, thereby increasing elastic potential energy of the elastic reset member; and the elastic reset member is configured to drive the piston to do reset movement when released, so that the liquid in the pump cylinder flows through the pump out hole to the nozzle for spraying.

13. The water gun according to claim 12, wherein the gear transmission group comprises a driving gear, a plurality of transmission gears, an intermittent gear, and a rack; the motor is in driving connection to the driving gear; the transmission gears are meshed in sequence; the driving gear is meshed with one of the transmission gears; the intermittent gear is coaxially connected to another transmission gear; the rack is connected to the piston, and the intermittent gear is intermittently meshed with the rack; when the intermittent gear is meshed with the rack, the driving gear is driven by the motor to drive the transmission gears to rotate, thereby driving the intermittent gear to rotate and driving the piston to move through the rack; and when the intermittent gear is separated from the rack, the elastic reset member is released to drive the piston to be reset and move.

14. The water gun according to claim 1, wherein a mounting slot is provided in the gun body; a liquid guide head is arranged in the mounting slot; the liquid guide head is communicated to the water inlet hole; the liquid storage device comprises a drum and a connector; a liquid chamber is provided inside the drum; the connector is arranged on the drum; the connector is communicated to the liquid chamber; and the drum is detachably inserted into the mounting slot, so that the connector is communicated to the liquid guide head.
